Taiyang Xu

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Current position

Université catholique de Louvain, Louvain-la-Neuve, Belgium — 08/2026 – now
Institut de Recherche en Mathématique et Physique
Postdoctoral Researcher (Mentor: Prof. Christophe Charlier)
Member of group *Random matrices, tilings, and water waves*

Past positions

Fudan University, Shanghai, China — 07/2024 – 06/2026
Department of Mathematics
Postdoctoral Fellow (Mentor: Prof. Lun Zhang)

Education

Fudan University, Shanghai, China — 09/2019 – 06/2024
Ph.D. in Mathematics, supervisor Prof. Engui Fan
Thesis title: "*On the long-time asymptotics of the local and nonlocal mKdV equation under the nonzero background*"

China University of Mining and Technology, Xuzhou, China — 09/2015 – 06/2019
B.Sc. in Mathematics, Distinguished Honor
Thesis title: "*Inverse scattering theory and integrability on several kinds of nonlinear evolution equations*"

Research interests

Integrable ODEs/PDEs, Random matrix theory, Determinantal point processes, Orthogonal polynomials, Asymptotic analysis, Riemann-Hilbert problems, Special functions ...

Publications and preprints

Listed in reverse chronological order of first public availability.

10. Moment generating function of the tacnode process
    arXiv:2606.17771
9. Critical asymptotics of the Camassa-Holm equation in the small dispersion limit at the point of gradient catastrophe (with Yiling Yang and Lun Zhang)
    available upon request
8. Confluent hypergeometric kernel determinant on multiple large intervals (with Lun Zhang and Zhengyang Zhao)
    arXiv:2508.10463
7. Transient asymptotics of the modified Camassa-Holm equation (with Yiling Yang and Lun Zhang)
    Journal of the London Mathematical Society, 110 (2024), e12967. / arXiv:2308.06950
6. Painlevé transcendents in the defocusing mKdV equation with non-zero boundary conditions (with Engui Fan and Zhaoyu Wang)
    Communications in Mathematical Physics, 406 (2025), 181. / arXiv:2306.07073
5. Large-time asymptotics to the focusing nonlocal modified Kortweg-de Vries equation with step-like boundary conditions (with Engui Fan)
    Studies in Applied Mathematics, 150 (2023), 1217–1273. / arXiv:2208.01268
4. On the large-time asymptotics of the defocusing mKdV equation with step-like initial data (with Yidan Zhang)
    Mathematical Physics, Analysis and Geometry, 29 (2026), 38. / arXiv:2204.01299
3. On the Cauchy problem of defocusing mKdV equation with finite density initial data: long-time asymptotics in soliton-less regions (with Engui Fan and Zechuan Zhang)
    Journal of Differential Equations, 372 (2023), 55–122. / arXiv:2108.06284
2. Soliton resolution and asymptotic stability of N-soliton solutions for the defocusing mKdV equation with a non-vanishing background (with Engui Fan and Zechuan Zhang)
    Physica D: Nonlinear Phenomena, 472 (2025), 134526. / arXiv:2108.03650

1. Riemann-Hilbert approach for multisoliton solutions of generalized coupled fourth-order nonlinear Schrödinger equations (with Weiqi Peng and Shoufu Tian)
Mathematical Methods in the Applied Sciences, 43 (2020), 865–880.

Grants

– Shanghai Post-doctoral Excellence Program (Grant No. 2024100) 07/2024 – 06/2026
Project: Riemann-Hilbert method for several asymptotic problems related to universality from integrable systems and random matrix theory
Role: Principal Investigator

– China Postdoctoral Science Foundation (Grant No. 2024M760480) 07/2024 – 06/2026
Project: Semiclassical asymptotics and universality for nonlinear integrable shallow water wave systems
Role: Principal Investigator

Teaching

09/2019 – 07/2026 @Fudan

– Fall, 2025: Recitation & Tutorial Instructor of Mathematical Analysis III (MATH130001.04).
– Spring, 2024: TA of Methods of Asymptotic Analysis (MATH630117).
– Fall, 2021: TA of Calculus A (MATH120021.02).
– Spring, 2020: TA of Calculus B (MATH120004.01) (Online).
– Fall, 2019: TA of Calculus B (MATH120003.01).

Research talks

– "*Large gap asymptotics for the confluent hypergeometric kernel determinant*" (invited), 2nd meeting of the FNRS contact group devoted to probability theory, Université libre de Bruxelles, Brussel, Belgium, 15–16 Oct, 2026.
– "*On the Fredholm determinants*" (invited), Seminar @Jiangsu Normal University, Xuzhou, China, 23 Jun, 2026.
– "*Dubrovin's conjecture for the Camassa-Holm equation*" (invited), 3rd Workshop on Integrable Systems and Random Matrix Theory 2026, Dongguan, China, 4–17 Jan, 2026.
– "*Confluent hypergeometric kernel determinant on multiple large intervals*" (invited), Bielefeld-Melbourne-Seoul Online Random Matrix Seminar, 15th Oct, 2025.
– "*Confluent hypergeometric kernel determinant on multiple large intervals*" (contributed), 5th ZiF Summer School "Randomness in Physics and Mathematics: From Thermalisation in Quantum Systems to Random Matrices", Universität Bielefeld, Bielefeld, Germany, 25 Aug–6 Sep, 2025.
– "*An introduction to Dubrovin's universality conjecture for integrable PDEs, and some progress on the Camassa-Holm equation*" (invited), Seminar @UCLouvain, Louvain-la-Neuve, Belgium, 21 Aug, 2025.
– "*Transient asymptotics of the modified Camassa-Holm equation*" (invited), The 2nd Workshop on Integrable Systems and Random Matrix Theory, Dongguan, China, 5–17 Jan, 2025.
– "*Integrable PDEs with nonzero boundary conditions: large-time asymptotics*" (contributed), The 15th Hemudu Forum on Integrable Systems, Ningbo, China, 24–26 Nov, 2023.

Other presentations

Outreach talks

– "*Some asymptotic problems in mathematical physics*" (invited), Shanghai Institute of Technical Physics, Shanghai, China, 29th Apr, 2025.

12 **presentations** (including reading reports) given at Fudan Integrable Systems and Random Matrix Theory Seminar.

Academic visits

- 22/06/2026 – 25/06/2026, Jiangsu Normal University, China. (Host: Dong Yao)
- 18/08/2025 – 24/08/2025, UCLouvain, Belgium. (Host: Christophe Charlier)
- 31/03/2025 – 11/04/2025, Chongqing University, China. (Host: Yiling Yang)

(I greatly appreciate for their warm hospitality)

Service

Co-organizer of the following activities

- (with Lun Zhang) Weekly RMT seminar @Fudan, 09/2022 – 06/2026.
- (with Junwen Liu and Lun Zhang) Mini-workshop on Random Matrices, Fudan University, Shanghai, China, 4 June, 2026.
- (with Engui Fan, Zhaoyu Wang, Yiling Yang and Lun Zhang) Workshop on Integrable Systems, Fudan University, Shanghai, China, 28 Nov, 2025.
- (with Lun Zhang) Mini-workshop on Asymptotic Analysis, Fudan University, Shanghai, China, 5–9 Jun, 2025.

Referee services provided for the following journals

- Analysis and Applications
- Journal of Nonlinear Mathematical Physics
- Qualitative Theory of Dynamical Systems
- Studies in Applied Mathematics

Other service

- zbMATH Open Reviewer

Scholarships and awards

09/2019 – 06/2024 @Fudan (Doctorate)

- Shanghai Outstanding Graduate (Graduation with Honors), 2024.
- Huatai Securities Science and Technology Scholarship, 2023.
- Pacific Insurance Company Scholarship, 2022.
- Outstanding Doctoral Candidate Scholarship, Fudan University, 2021.
- Fudan University Doctoral Scholarship, 2019–2023.

09/2015 – 06/2019 @CUMT (Undergraduate)

- Outstanding Undergraduates in China University of Mining and Technology, 2019.

Status

China – citizen, **Married**.

Languages

- Chinese (native), The Qianwei dialect of Sichuanese (native)
- English

Computer skills

LaTeX, Mathematica, Matlab, HTML, C++, Javascript

Tuesday 20th October, 2026